

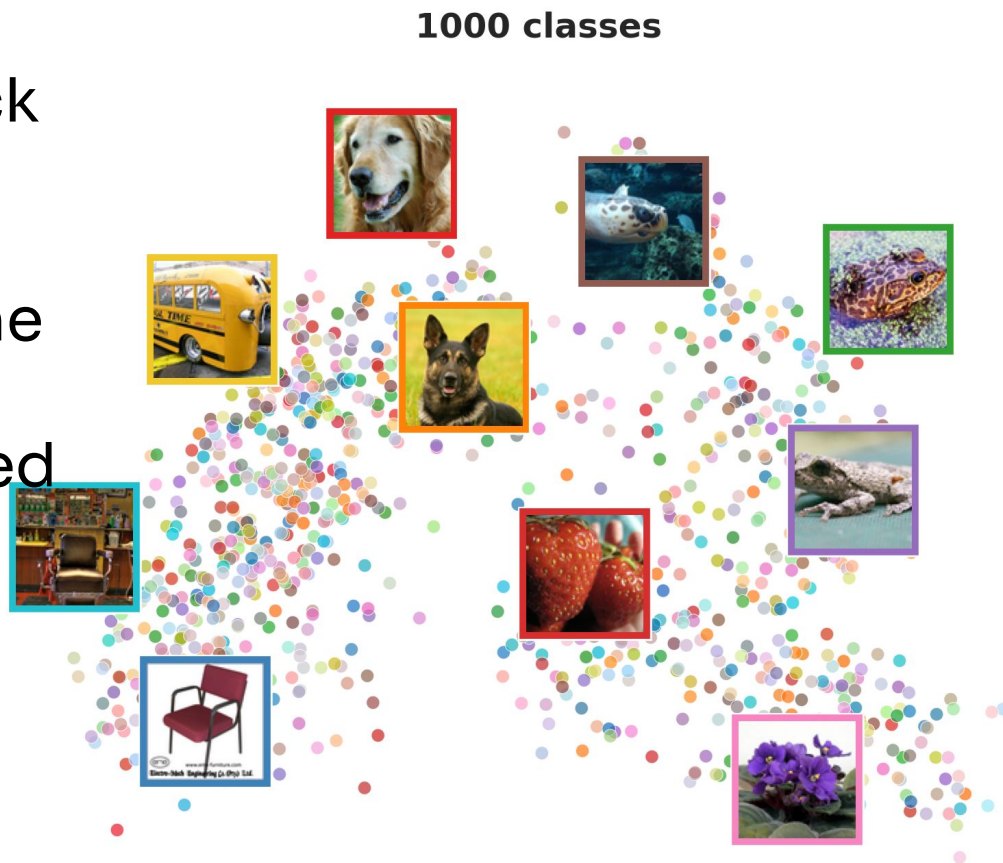
An extremely coarse feedback signal for learning human-aligned visual representations

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Bonner Lab
Cognitive Science

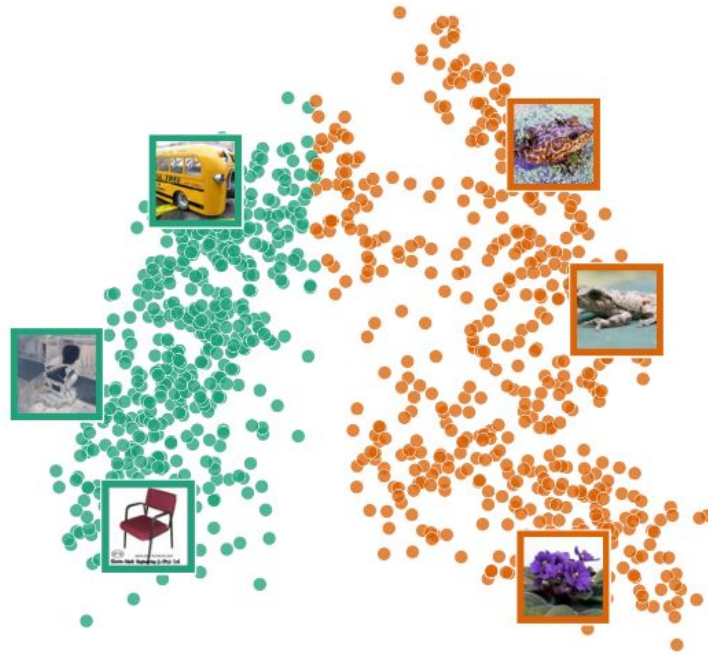
18th May 2026
VSS



Deep learning has been moving to finer feedback signals. (early, obj classification); what is “fine-grained”; that is the central approach; (all modern DNNs are trained on fine-grained tasks!); eg. self-supervised; (talía)

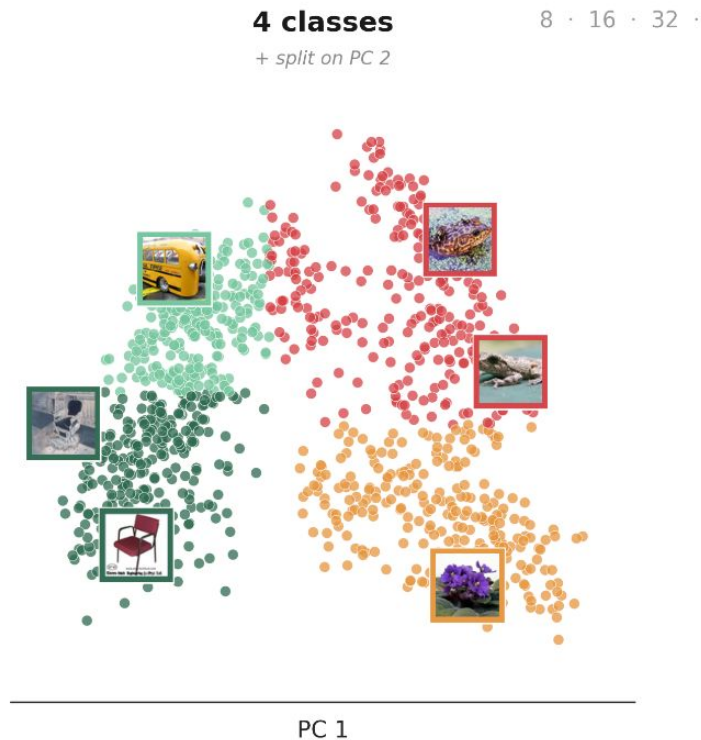
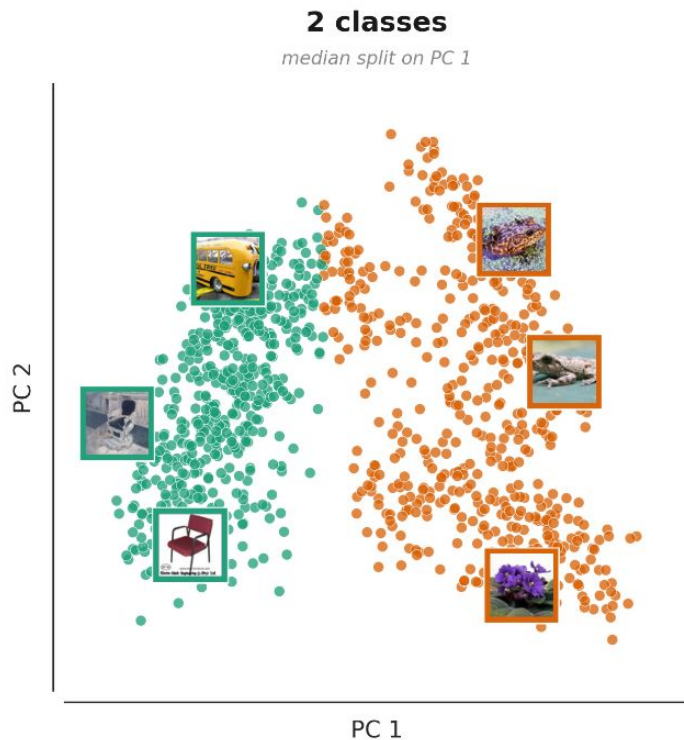


Infants don't receive fine-grained feedback signals

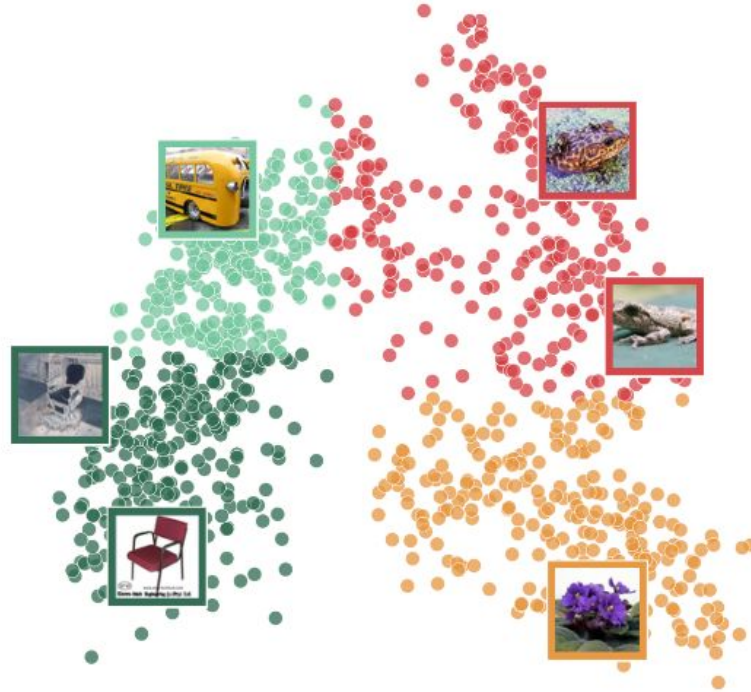
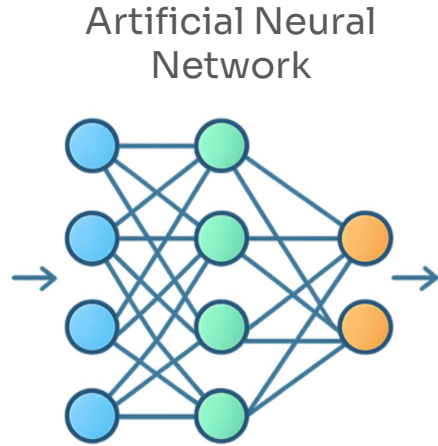


2 classes

Binary split based on the i-th Principal Component

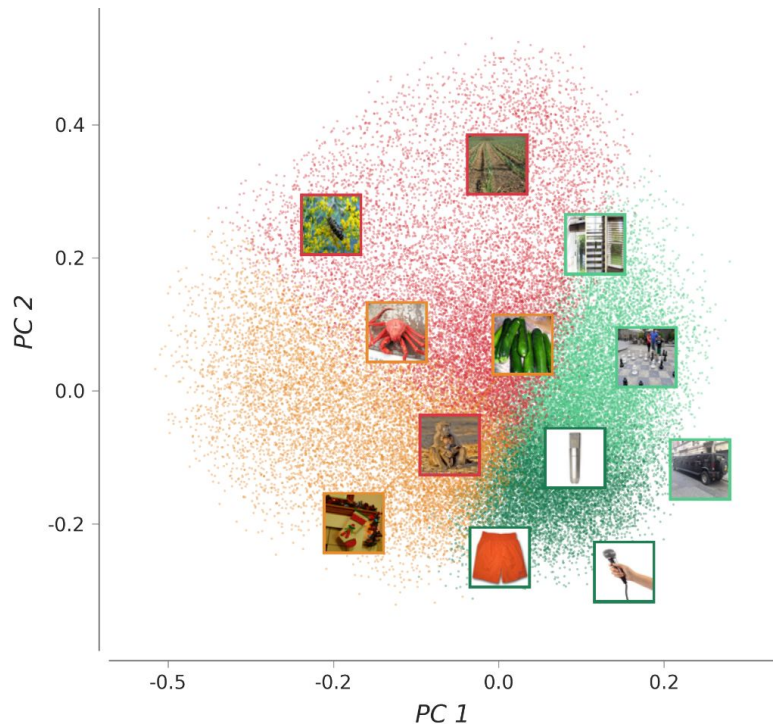


Train neural network from scratch on coarse classes

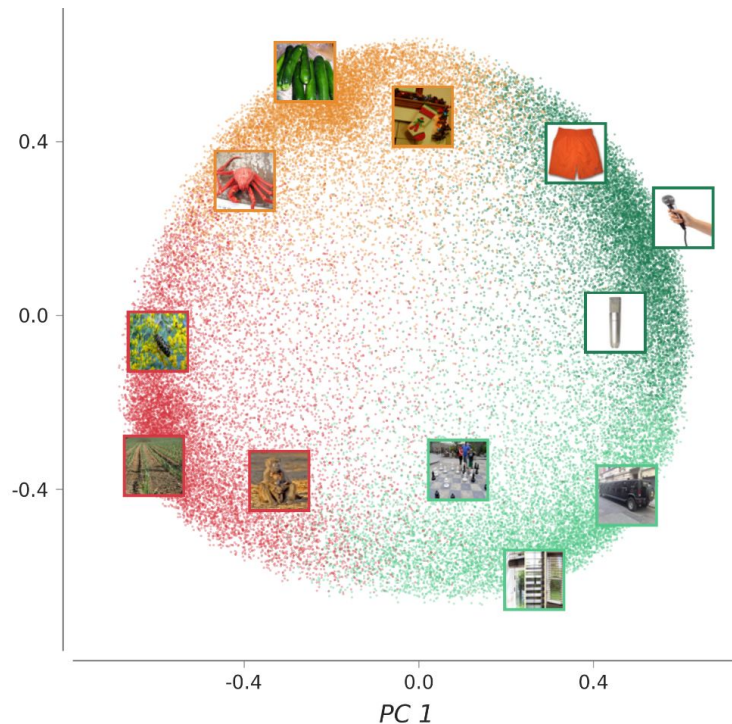


ANNs learn more categorical internal representations

1000 classes

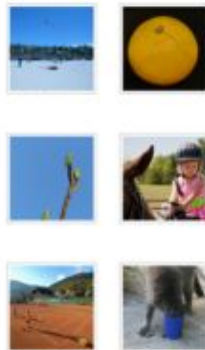


4 classes



d

73,000
natural scenes

**NSD**

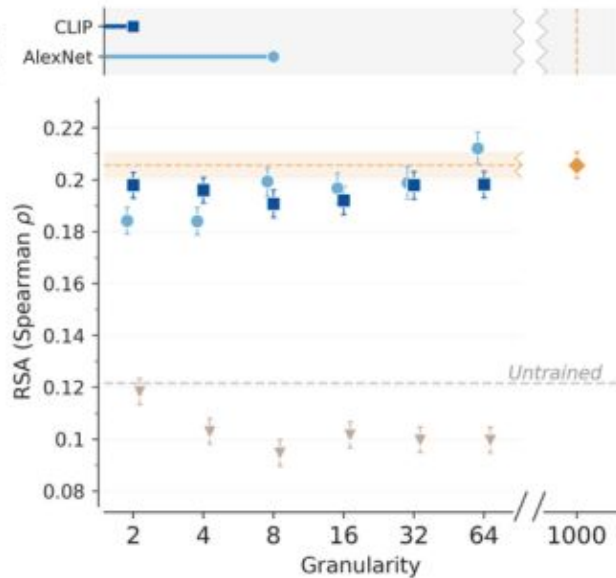
8 human subjects
ventral cortex



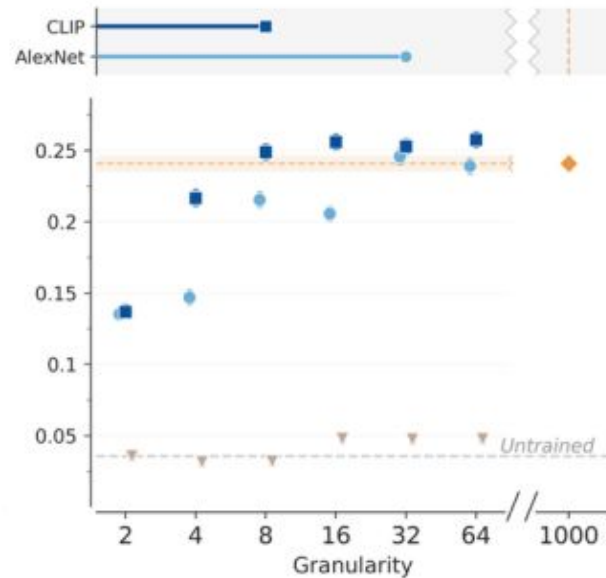
Human
7T fMRI

e

Early visual stream

**f**

Ventral visual stream



THINGS behavior database

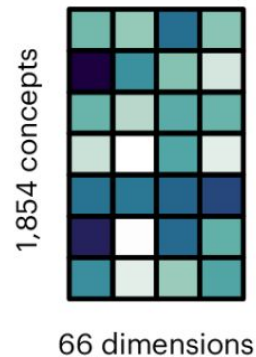
Visual similarity



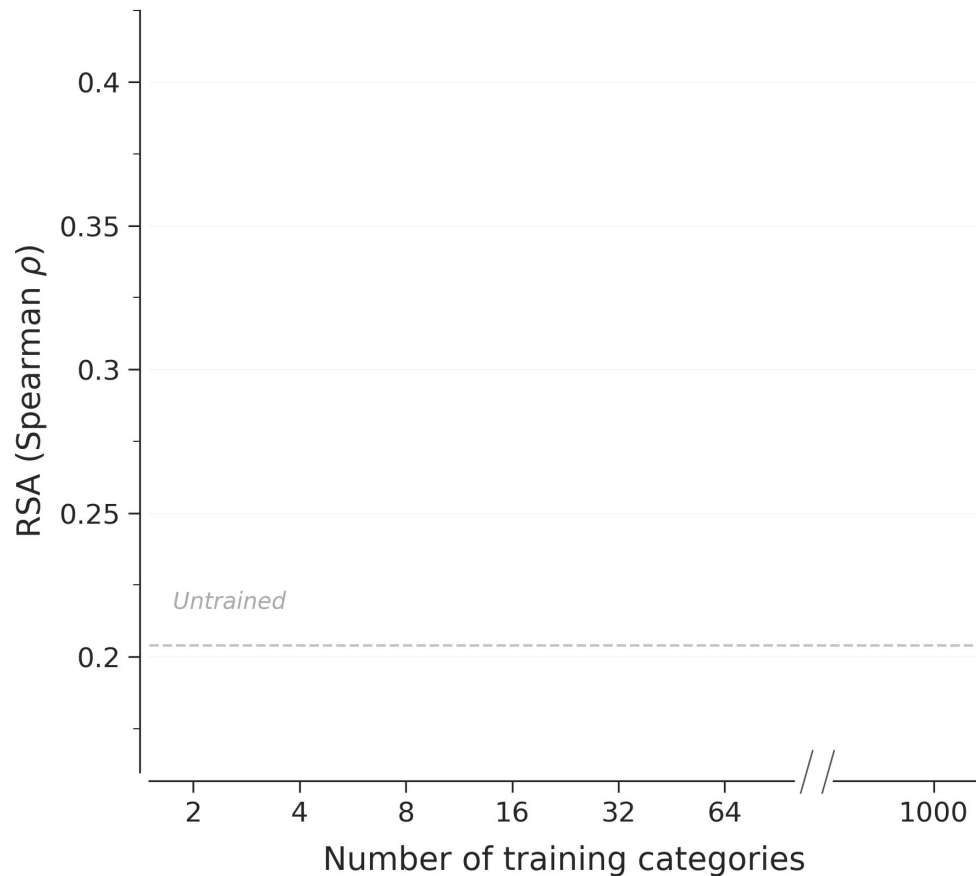
Behaviour



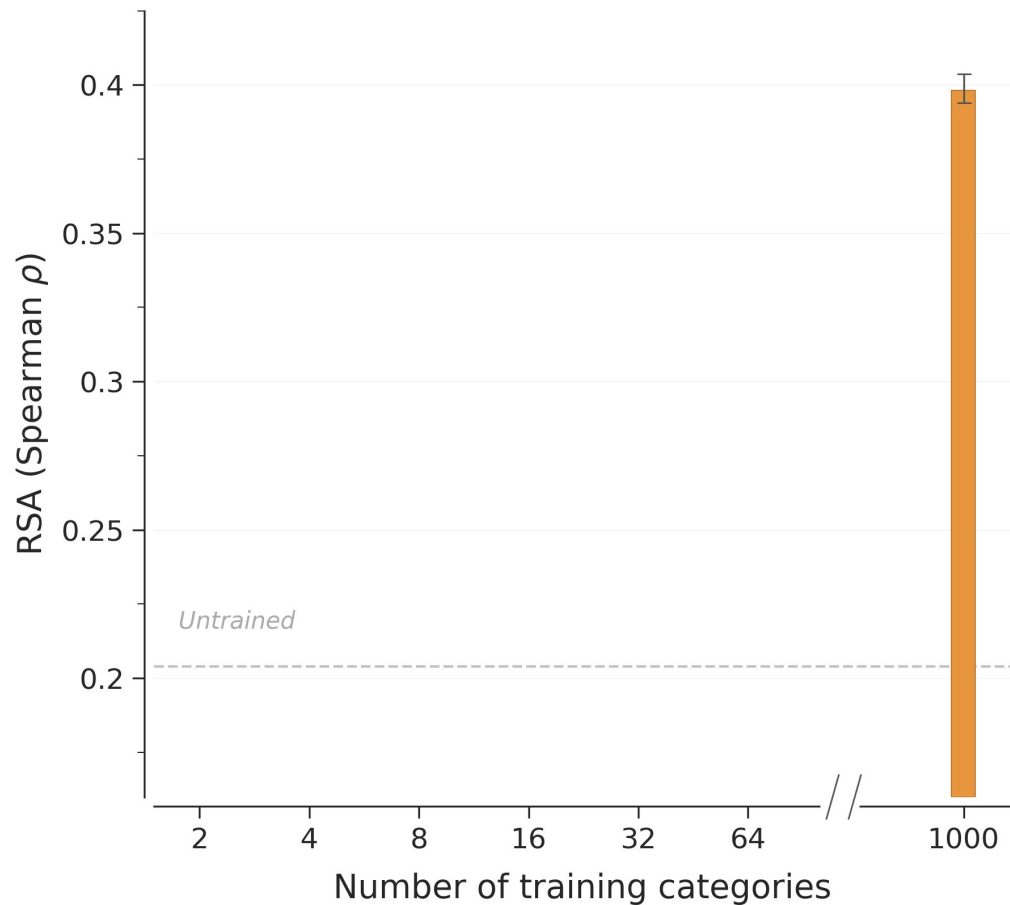
Similarity
embedding



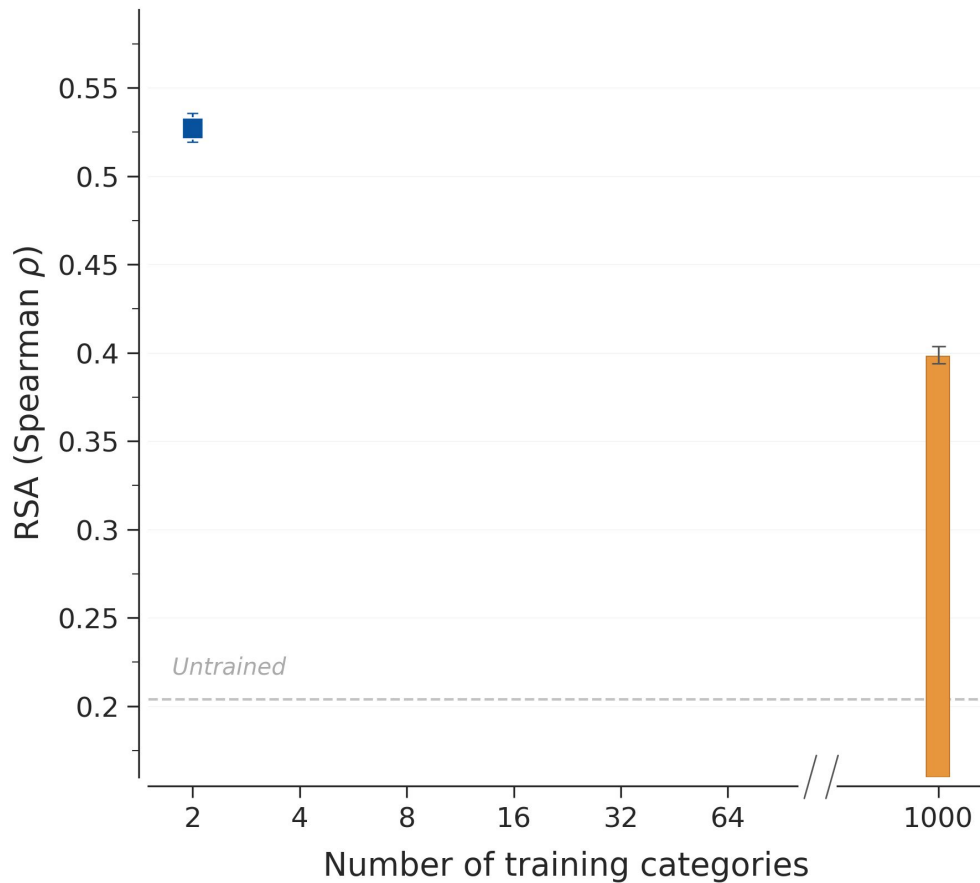
Alignment with human similarity measurements



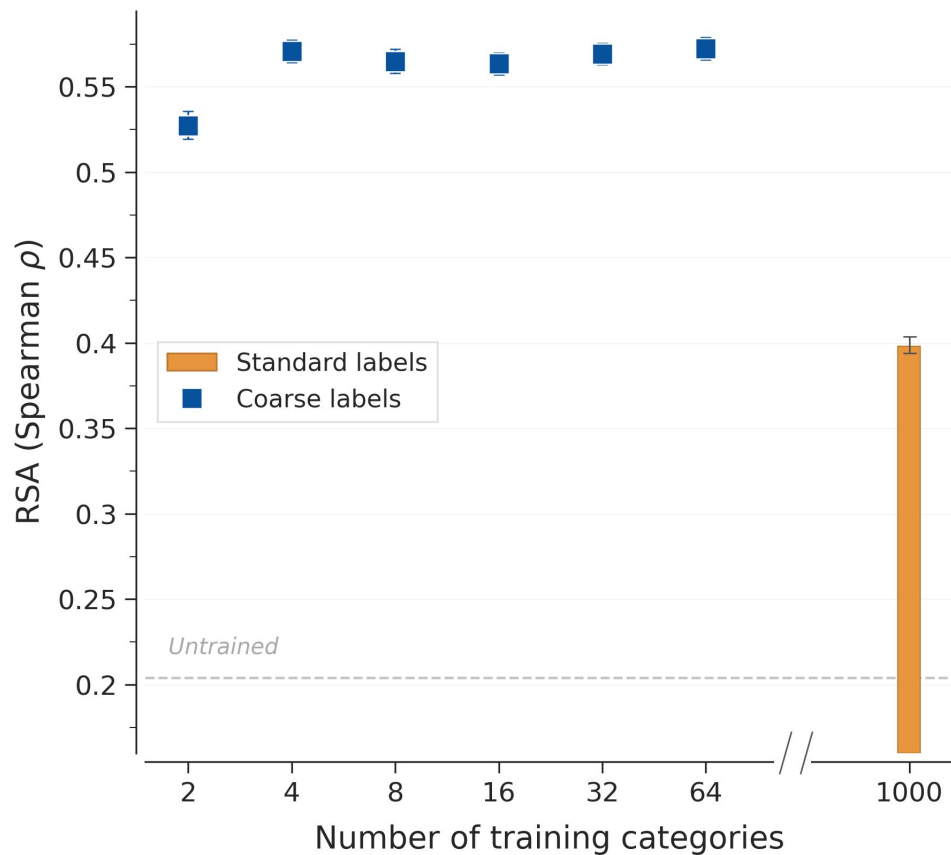
Alignment with human similarity measurements



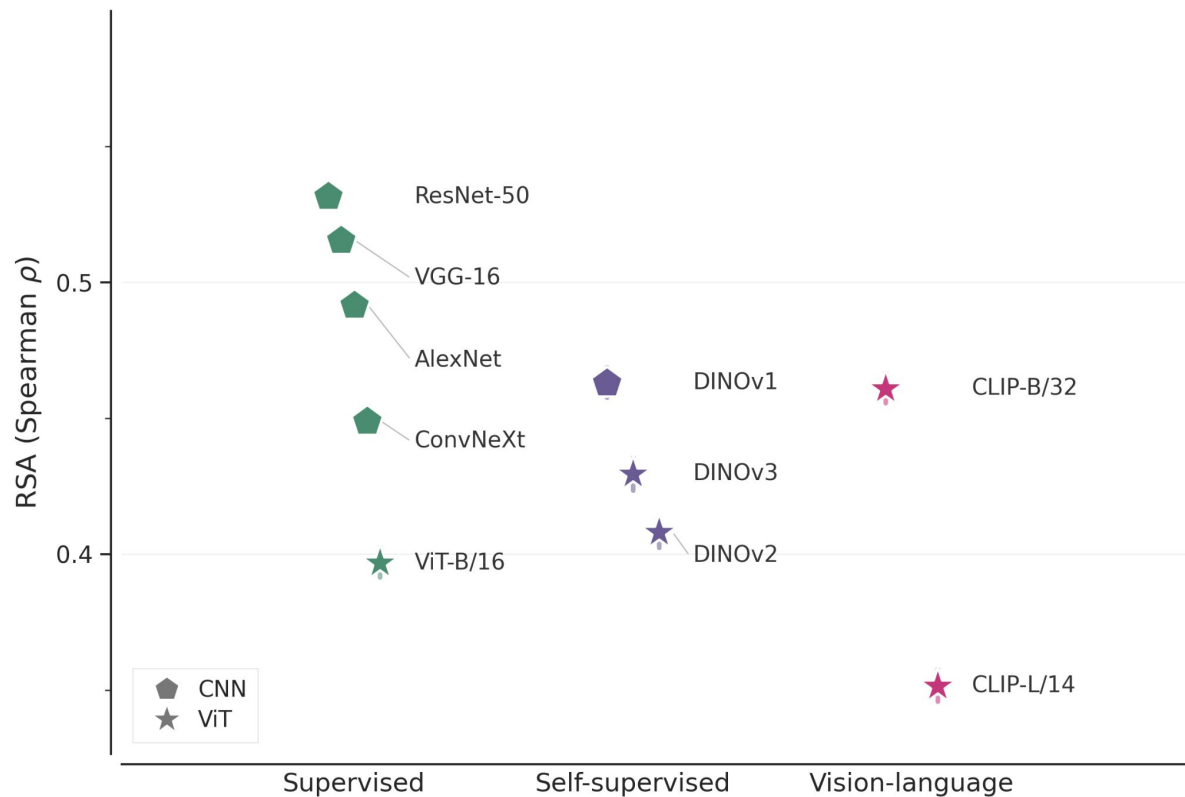
Alignment with human similarity measurements



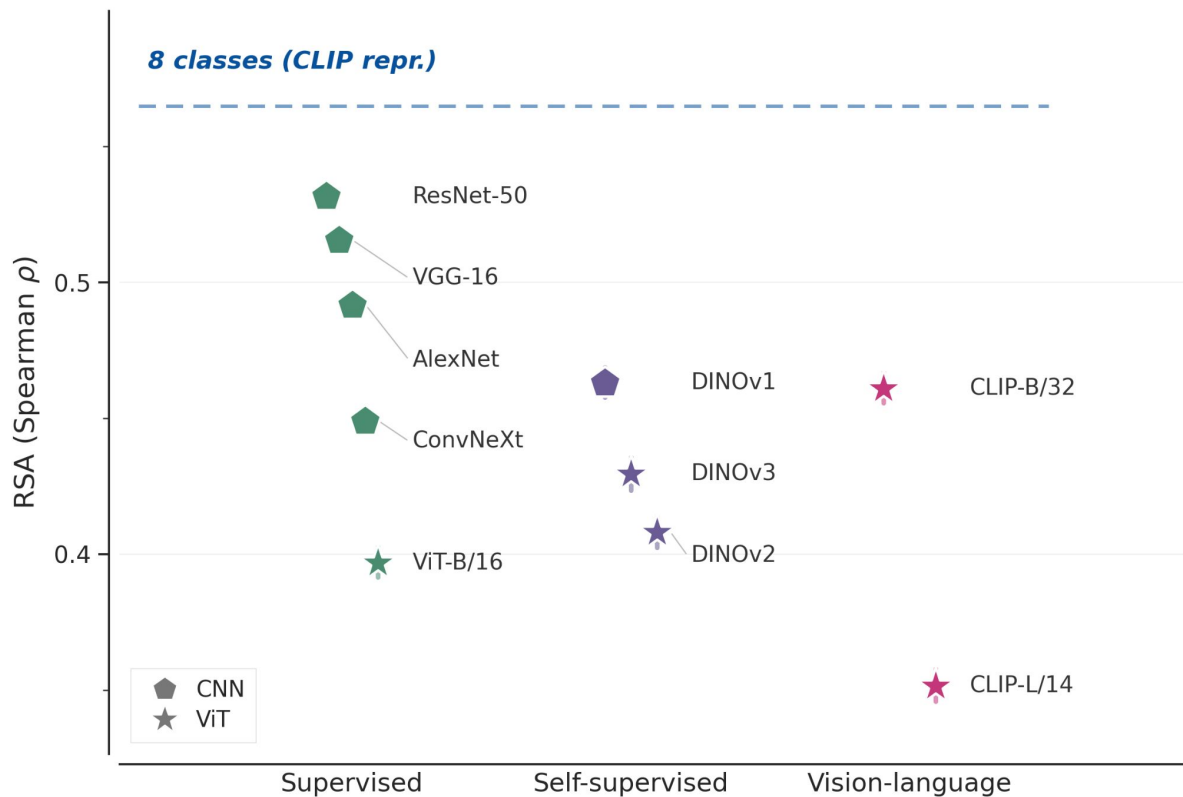
Alignment with human similarity measurements



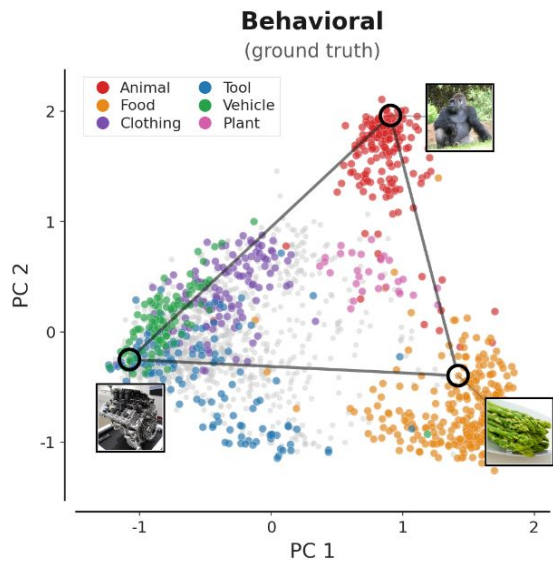
Comparison with pre-trained models



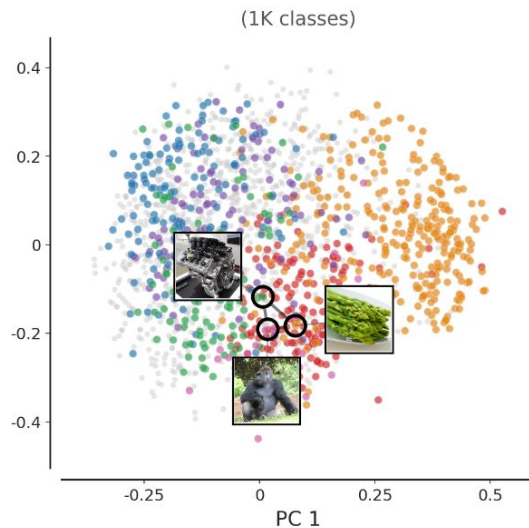
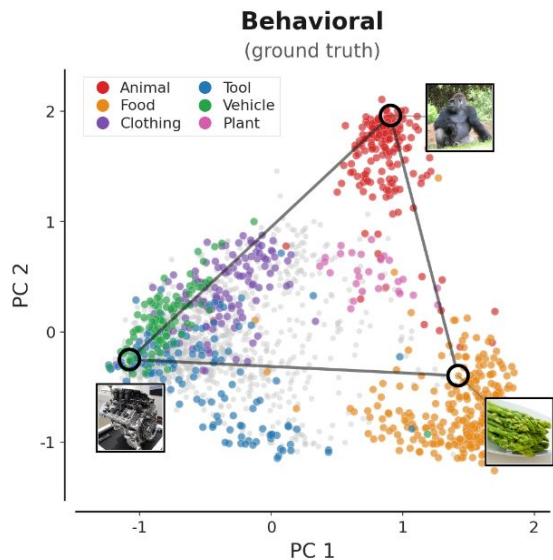
Comparison with pre-trained models



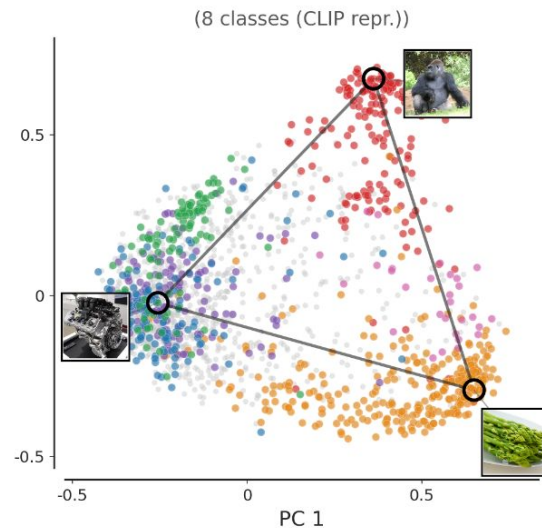
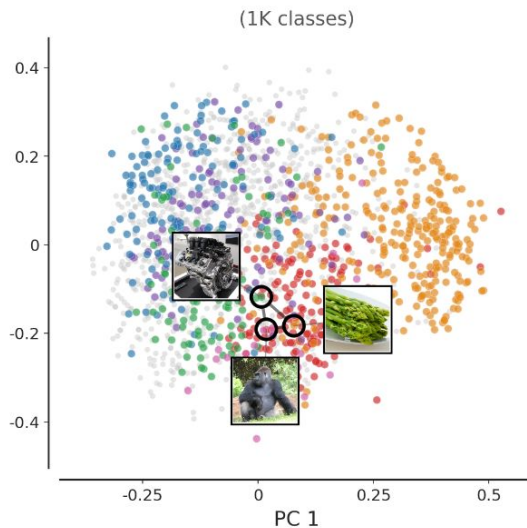
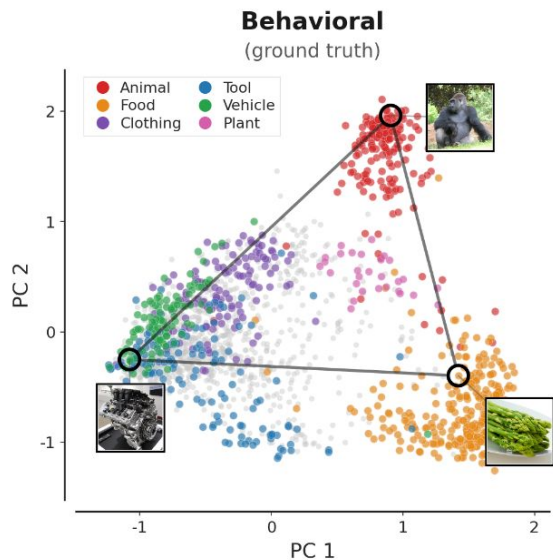
Comparing 2D representational space



Comparing 2D representational space



Comparing 2D representational space



We find a way to create a coarse learning signal from data to make artificial neural networks more human-aligned.

Team

Jump to [Lab](#), [Alumni](#), [Collaborators](#), [Former students](#), [Administrative support](#)



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Thank you & questions!

Lab



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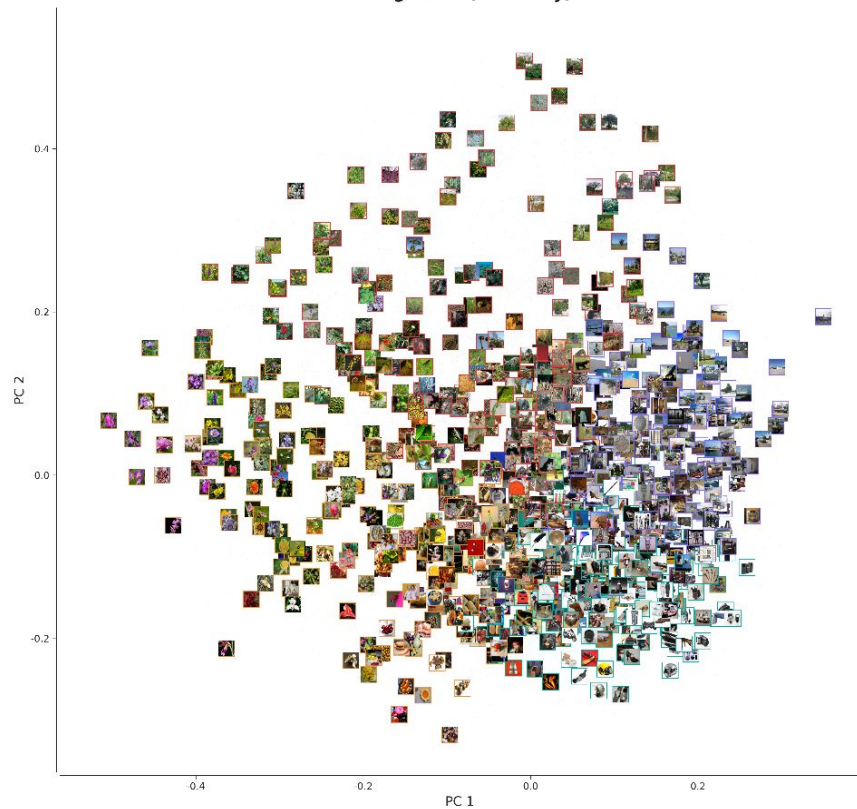


Ananya Passi

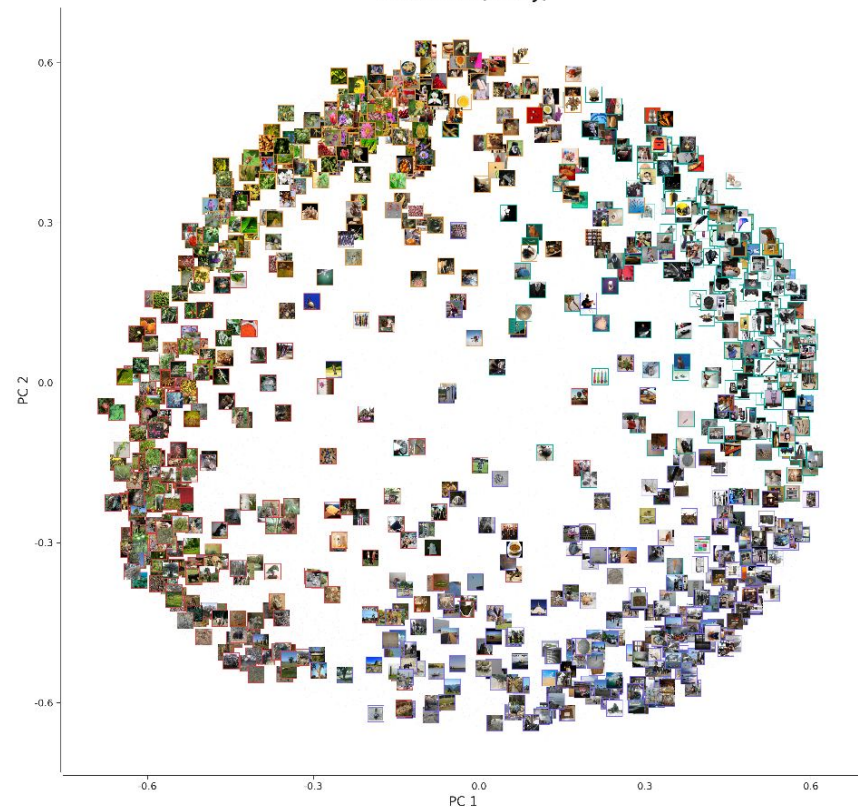
PhD Student, Cognitive Science



a Fine-grained (1000-way)

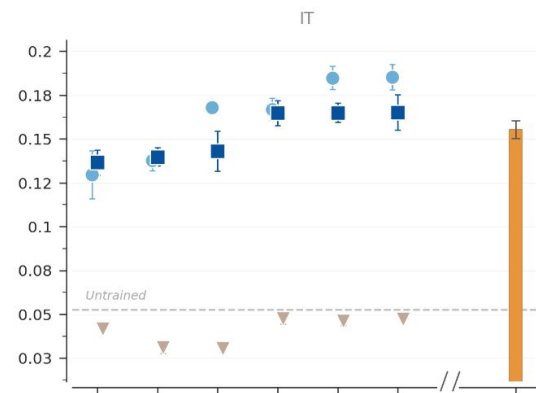
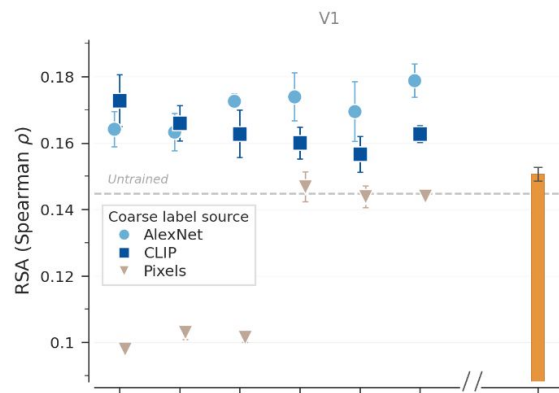


b Coarsened (4-way)

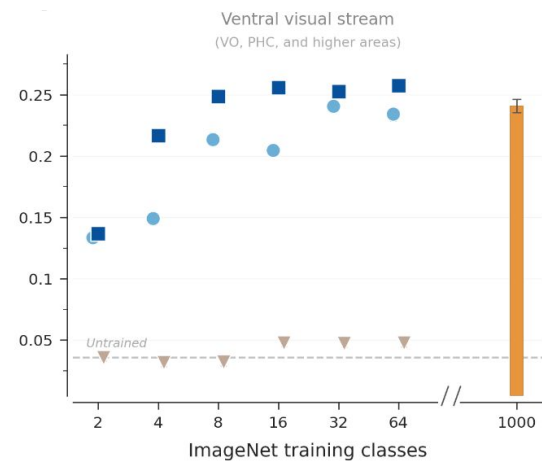
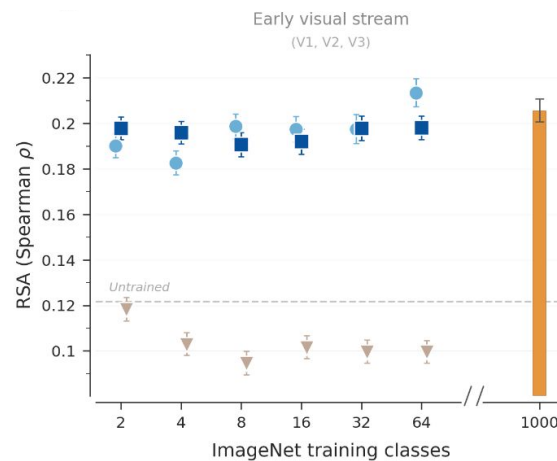


■ Class 0 ■ Class 1 ■ Class 2 ■ Class 3

TVSD



NSD



Model comparisons

